

Role of Python, SQL and Power BI in Financial Data Analytics

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Abstract

This paper examines the role of Python, SQL, and Power BI in financial data analytics. Python libraries such as Pandas, NumPy, and Matplotlib are used for data cleaning, analysis, and visualization, while SQL and MySQL are used for data storage and querying. Power BI is used to create interactive dashboards and reports. The study explains how these technologies work together to process financial data and generate meaningful insights. The proposed workflow demonstrates how database management, data analysis, visualization, and business intelligence tools can be integrated into a single financial analytics process.

Index Terms - Python, Pandas, NumPy, Matplotlib, SQL, MySQL, Power BI, Financial Data Analytics

I. Introduction

Financial data is generated in large volumes from mutual funds, stock exchanges, banking systems, and other financial institutions. Analysing this data manually can be time-consuming and may make it difficult to identify important trends and patterns.

Data analytics technologies provide tools for collecting, storing, cleaning, analysing, and visualizing financial information. Python is widely used for data processing and analysis, SQL is used for database management and querying, and Power BI is used for interactive reporting and dashboard creation.

This paper studies the role of Python, SQL, and Power BI in financial data analytics using a practical financial dataset. The workflow focuses on mutual fund and direct equity data and demonstrates how different technologies can work together to produce useful analytical results.

II. Objectives

The main objectives of this study are:

- 1) To understand the role of Python in financial data analysis.
- 2) To use Pandas and NumPy for data processing and numerical analysis.
- 3) To use SQL and MySQL for storing and querying financial data.
- 4) To connect Python with the MySQL database.
- 5) To use Matplotlib for financial data visualization.
- 6) To create an interactive financial dashboard using Power BI.
- 7) To understand how different technologies can be integrated into a financial analytics workflow.
- 8) To generate meaningful insights from mutual fund and direct equity data.

III. Python in Financial data analytics

Python is a programming language commonly used for data analysis, automation, and visualization. It provides several libraries that simplify financial data processing.

In this project, Python is used for loading datasets, checking data quality, performing calculations, grouping data, and creating visualizations.

The major Python libraries used in this project are Pandas, NumPy, and Matplotlib. Python provides a simple environment where different data analysis tasks can be performed efficiently.

IV. Pandas for data processing

Pandas is a Python library used for data manipulation and analysis. It provides DataFrame structures that make it easier to work with tabular financial data.

In this project, Pandas is used to read CSV files, check dataset shape and columns, identify missing values, remove unwanted rows, remove duplicate records, group data by year and month, calculate average values, identify top mutual fund schemes, and prepare data for visualization.

For example, yearly average AUM and yearly average net inflow are calculated using Pandas groupby operations. Pandas helps convert raw financial data into a structured format suitable for analysis.

V. NumPy for numerical analysis

NumPy is a Python library used for numerical and mathematical operations. It provides efficient support for numerical calculations and arrays.

In financial data analysis, numerical operations are required for calculating averages, indexes, and comparing values. In this project, NumPy is included as part of the Python data analysis environment and supports numerical processing where required.

The use of NumPy along with Pandas provides a suitable environment for handling numerical financial data.

VI. SQL and MySQL for Database Management

SQL stands for Structured Query Language. It is used to store, retrieve, filter, and analyse data in relational databases.

MySQL is used in this project as the database management system. Financial datasets are stored in database tables such as the Mutual Fund table and Direct Equity table.

SQL queries are used to retrieve records and perform operations such as SELECT, WHERE, GROUP BY, ORDER BY, aggregate functions, and JOIN operations. Using SQL allows financial data to be stored systematically and retrieved when required.

VII. Python – Database Integration

Python can be connected to MySQL to retrieve database information directly into a Python environment.

In this project, the Python database connection is used to connect Python with MySQL, access the project database, retrieve

mutual fund data, retrieve direct equity data, load the retrieved data into Pandas DataFrames, and use the data for further analysis.

This integration reduces the need to manually export data from the database every time analysis is performed. The combination of SQL and Python creates a connection between database management and data analysis.

VIII. Matplotlib for data visualization

Matplotlib is a Python library used to create charts and graphs. In this project, Matplotlib is used to visualize year-wise average AUM, year-wise average net inflow, month-wise average AUM, direct equity settlement value, BSE vs NSE equity value, mutual fund vs direct equity trend, top mutual fund schemes, and positive vs negative net inflow.

Different chart types such as line charts, bar charts, and horizontal bar charts are used according to the analysis requirement. Visualizations make numerical financial data easier to understand and help identify changes and trends.

IX. Power BI for interactive dashboards

Power BI is a business intelligence and visualization tool used to create interactive dashboards and reports.

In this project, Power BI is used to present the analysed financial data in an interactive format. The dashboard contains Total AUM, Average Net Inflow, Average Direct Equity Value, Total Records, Mutual Fund vs Direct Equity Trend, Mutual Fund Net Inflow Trend, BSE vs NSE Equity Value Trend, BSE vs NSE Equity Value Share, and Mutual Fund Folio Trend.

A year slicer is also used to allow users to filter the dashboard by year. Power BI helps convert analytical results into an easy-to-understand business dashboard.

X. Integrated financial data analytics workflow

The technologies used in this project work together as a complete data analytics workflow.

Data Sources → CSV Data → MySQL Database → SQL Queries → Python/Pandas → Data Cleaning and Analysis → Matplotlib Visualization → Power BI Dashboard → Insights.

Each technology performs a specific role. Python is used for analysis and processing, Pandas for data cleaning and manipulation, NumPy for numerical operations, MySQL for data storage, SQL for querying, Matplotlib for visualization, and Power BI for interactive dashboards.

This integrated approach allows financial data to move from raw data collection to final visualization and reporting.

XI. Applications of financial data analysis

The workflow can be applied to different financial datasets such as mutual funds, stock market data, banking data, and investment records.

For this project, mutual fund and direct equity datasets are analysed. The analysis includes year-wise AUM analysis, net inflow analysis, monthly AUM analysis, direct equity settlement value analysis, BSE and NSE comparison, mutual fund and direct equity trend

comparison, top mutual fund schemes by average AUM, and positive and negative net inflow analysis. The results are presented through Python visualizations and a Power BI dashboard.

XII. Advantages of the technology based approach

The use of Python, SQL, and Power BI provides several advantages:

- 1) Structured Data Management: SQL and MySQL provide organized storage of financial data.
- 2) Efficient Data Processing: Pandas makes data cleaning and manipulation easier.
- 3) Numerical Analysis: NumPy supports numerical calculations.
- 4) Visualization: Matplotlib converts numerical results into understandable charts.
- 5) Interactive Reporting: Power BI provides interactive dashboards.
- 6) Reduced Manual Work: Database connectivity and Python reduce repetitive data handling.
- 7) Better Understanding: Visual representations make financial trends easier to interpret.

XIII. Limitations

The project also has some limitations:

- 1) The analysis depends on the quality and availability of the financial datasets.
- 2) The study uses selected financial indicators rather than all possible market indicators.
- 3) The analysis period is limited to the available dataset.
- 4) Power BI dashboards depend on the data loaded into the model.
- 5) The analysis provides historical insights and does not guarantee future financial performance.
- 6) Different financial data sources may use different definitions and measurement methods.

XIV. Role of each technology

Python is used as the main programming environment for financial data analysis. Pandas is used for data loading, cleaning, grouping, and manipulation. NumPy is used for numerical processing. SQL/MySQL is used for storing and retrieving financial data. Matplotlib is used to create charts for financial trend analysis. Power BI is used to create interactive dashboards and present analytical results.

Together, these technologies form a practical workflow for financial data analytics.

XV. Results and discussion

The implementation demonstrates that combining database technology, programming, and business intelligence tools can support financial data analysis.

Python-based analysis helps identify year-wise and month-wise trends in mutual fund data and direct equity data. SQL provides a structured method for storing and retrieving the datasets. Matplotlib provides graphical representations of the analysis, while Power BI presents the results through an interactive dashboard.

The project also demonstrates that different tools can be used at different stages of the analytics process instead of depending on a single technology. The integrated workflow provides a systematic approach from data storage to final visualization.

XVI. Conclusion

This study demonstrates the role of Python, SQL, and Power BI in financial data analytics. Python provides a flexible environment for data cleaning, processing, analysis, and visualization. Pandas and NumPy support data manipulation and numerical operations, while SQL and MySQL provide structured data storage and retrieval.

Matplotlib helps visualize financial trends, and Power BI provides an interactive dashboard for presenting the results.

The project demonstrates how these technologies can be integrated into a single workflow for analysing financial datasets. Such an approach can help students and analysts understand financial data more systematically and communicate analytical results through visual reports and dashboards.

References

- [1] W. McKinney, Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Jupyter, 3rd ed., O'Reilly Media, 2022.
- [2] Python Software Foundation, "Python Documentation." Available: <https://docs.python.org/3/>
- [3] MySQL, "MySQL Documentation." Available: <https://dev.mysql.com/doc/>
- [4] Microsoft, "Power BI Documentation." Available: <https://learn.microsoft.com/power-bi/>
- [5] Matplotlib Development Team, "Matplotlib Documentation." Available: <https://matplotlib.org/stable/>